

```

18th.py - C:\Users\thala\Downloads\FODS\18th.py (3.14.3)
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import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

df = pd.DataFrame({
    "Age": [23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56,
    "Fat": [9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 3
    34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7]
})

print(df.mean())
print(df.median())
print(df.std())

df.boxplot()
plt.show()

plt.scatter(df.Age, df.Fat)
plt.xlabel("Age")
plt.ylabel("Fat")
plt.show()

stats.probplot(df.Age, plot=plt)
plt.show()

```

```

Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:
\Users\thala\Downloads\FODS\18th.py =====
Age      46.444444
Fat      28.783333
dtype: float64
Age      51.0
Fat      30.7
dtype: float64
Age      13.218624
Fat       9.254395
dtype: float64

```

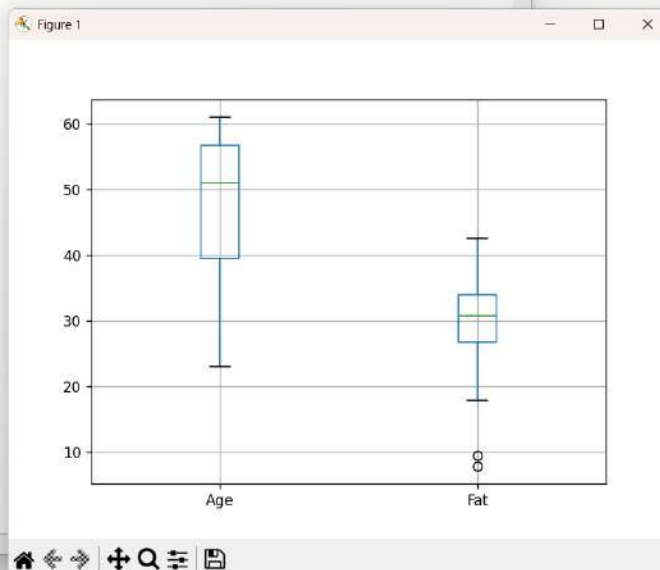
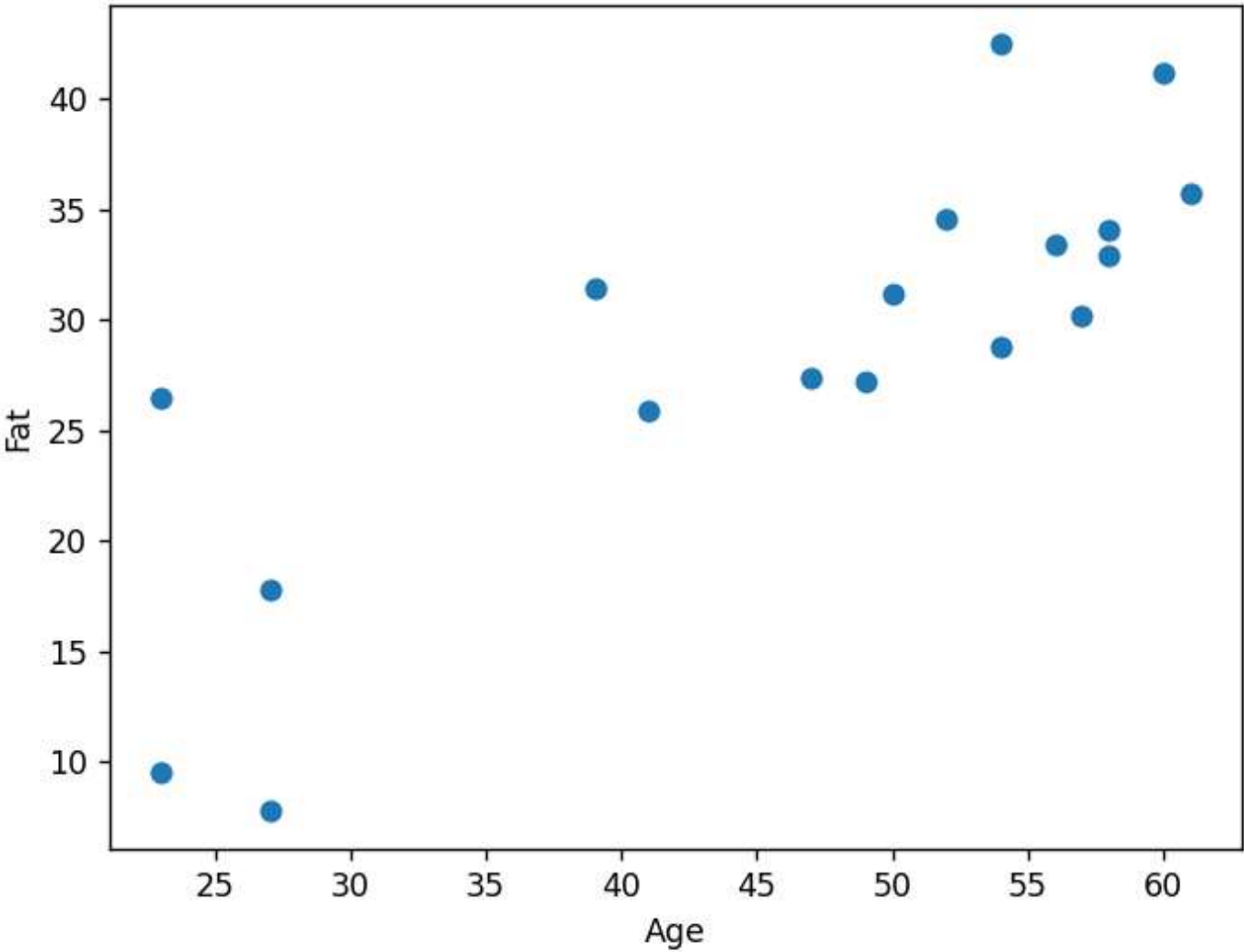




Figure 1



19th.py - C:\Users\thala\Downloads\FODS\19th.py (3.14.3)

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```
import pandas as pd
```

```
data = {  
    "Product": ["A", "B", "C", "D", "E", "F"],  
    "Quantity Sold": [100, 80, 120, 60, 90, 70],  
    "Unit Price": [200, 300, 150, 400, 250, 350]  
}
```

```
df = pd.DataFrame(data)
```

```
df["Total Sales"] = df["Quantity Sold"] * df["Unit Price"]  
df["Profit"] = df["Total Sales"] * 0.20
```

```
print(df.sort_values("Profit", ascending=False).head(5))
```

IDLE Shell 3.14.3

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```
>>>  
===== RESTART: C:  
\\Users\\thala\\Downloads\\FODS\\19th.py =====
```

```
Age      46.444444  
Fat      28.783333  
dtype: float64  
Age      51.0  
Fat      30.7  
dtype: float64  
Age      13.218624  
Fat       9.254395  
dtype: float64
```

```
===== RESTART: C:\Users\thala\Downloads\FODS\19th.py =====
```

	Product	Quantity Sold	Unit Price	Total Sales	Profit
5	F	70	350	24500	4900.0
1	B	80	300	24000	4800.0
3	D	60	400	24000	4800.0
4	E	90	250	22500	4500.0
0	A	100	200	20000	4000.0

```
>>>
```

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20th.py - C:\Users\thala\Downloads\FODS\20th.py (3.14.3)

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```
import pandas as pd
```

```
data = {
    "Customer ID": [1,2,3,4,5,6],
    "Age": [25,30,35,40,28,45],
    "Gender": ["M","F","M","F","M","F"],
    "Total Spending": [1200,700,300,1500,450,800]
}
```

```
df = pd.DataFrame(data)
```

```
def segment(x):
    if x >= 1000:
        return "High"
    elif x >= 500:
        return "Medium"
    else:
        return "Low"
```

```
df["Segment"] = df["Total Spending"].apply(segment)
```

```
print(df.groupby("Segment")["Age"].mean())
```

IDLE Shell 3.14.3

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Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
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```
>>> ===== RESTART: C:\Users\thala\Downloads\FODS\20th.py =====
```

```
Segment
High    32.5
Low     31.5
Medium  37.5
Name: Age, dtype: float64
```

```
>>>
```